



# IDBI INNOVATE 2026

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## Team Details

- a. Team name: **Looper**
- b. Team leader name: **Bhunesh Bansal**
- c. Problem Statement: **PS3 — Financial Health Score (AI-powered MSME Financial Health Card)**

## Brief about the idea

SaakhScore (साख — a trader's earned creditworthiness) is a banker-grade MSME Financial Health Card. It turns consented alternate data — GST, UPI, Account Aggregator bank statements, EPFO, sector, geography, vintage, and bureau status — into an explainable **0–100 score, A–E grade, risk band, indicative credit limit, underwriter memo, and borrower improvement plan.**

It is built for **New-to-Credit and New-to-Bank MSMEs** that traditional bureau-first underwriting rejects even when their real operating data is healthy — the rejected-to-approved reversal is visible in the first minute of the demo.

### 0–100 + A–E

score, grade, PD and policy kept  
separate

### EN + हिन्दी

plain-language reason codes

### Real outcomes

PD methodology proven on  
418,947 real loans

### 150/150 tests

live on Render · CI green

## Opportunities

- How different is it from any of the other existing ideas?
- How will it be able to solve the problem?
- USP of the proposed solution

### How it differs

Most teams and fintech demos stop at a score. SaakhScore exposes the full decision path: reason codes, exact Shapley attribution, tamper-evident audit trail, policy guardrails, source map, validation metrics, pilot KPIs, and a fairness monitor — all backend-verifiable, none mocked.

### How it solves the problem

It makes viable thin-file MSMEs visible to IDBI without turning model-assisted underwriting into a black box. Score, PD and lending policy stay separate; the cross-domain PD proxy can route a case to human review but can never auto-decline it.

### USP

The rejected-to-approved NTC reversal is visible in the first minute and backed by regulator-ready evidence — and the PD methodology is validated on real small-business default outcomes with a true later-in-time holdout, not on self-generated synthetic labels.

## List of features offered by the solution

**Five-pillar score:** Liquidity, Discipline, Momentum, Leverage, Digital Footprint

**Traditional bureau-only verdict vs alternate-data verdict** side by side

**Indicative working-capital limit** sized from spare EMI capacity, grade multiple as cap, full limit\_basis breakdown

**Native exact TreeSHAP** from a calibrated monotonic XGBoost champion; calibrated dependency-free logistic fallback

**Plain-language reason codes in English and Hindi**

- Underwriter memo + borrower improvement plan

**Policy guardrails:** EWS monitoring signals, counterparty-concentration limit, GST-vs-bank turnover reconciliation (both directions), bilingual next-best-action

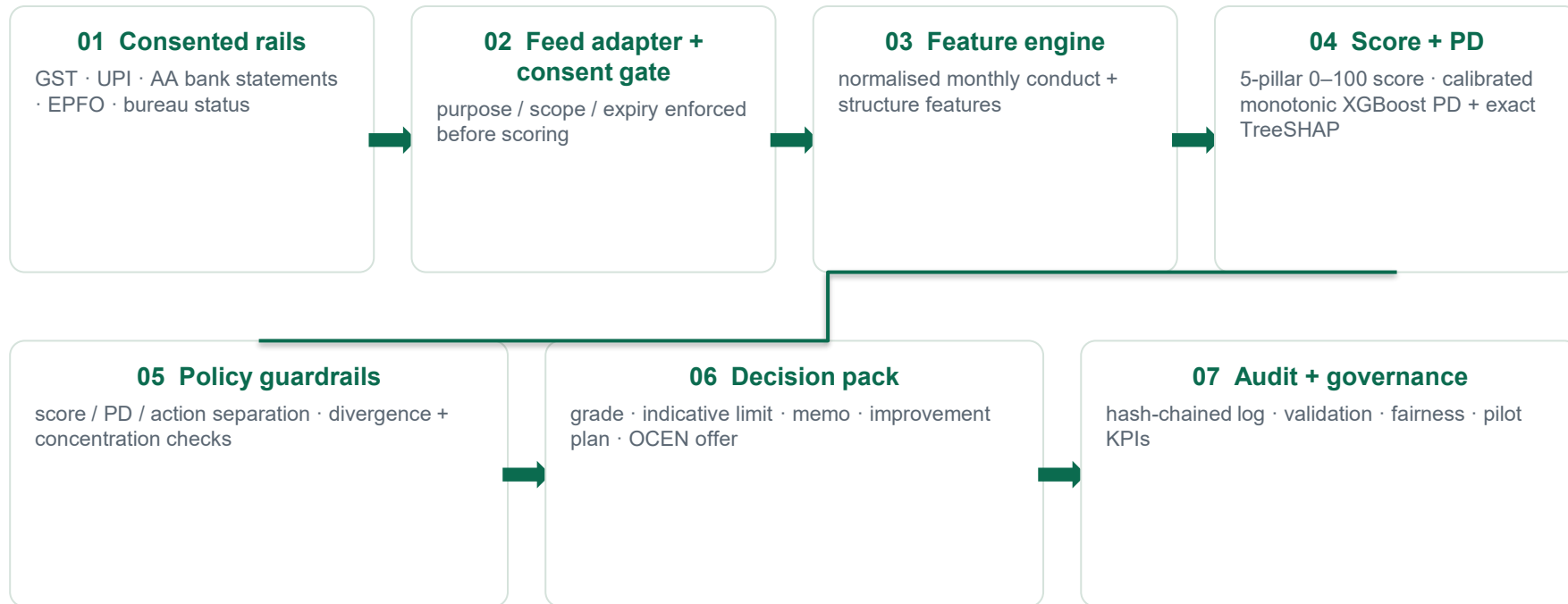
**Hash-chained, tamper-evident audit log**, reconstructable per decision

**IDBI sandbox-style AA/GST/UPI/EPFO feed endpoint** with enforced consent (purpose, scope, expiry)

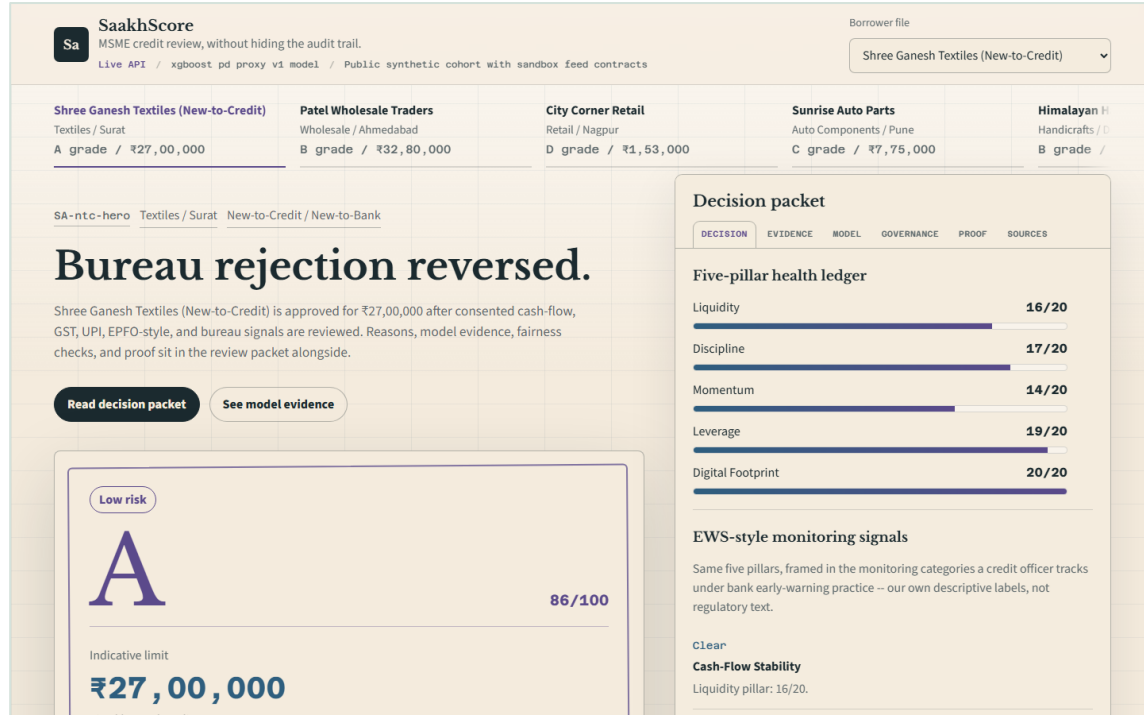
**OCEN 4.0-aligned offer artifact** + single-lever what-if re-score

- Portfolio impact, pilot metrics, validation report, governance endpoints
- Fairness view by sector, geography, vintage, gender (where available), bureau status

## Process flow diagram or Use-case diagram



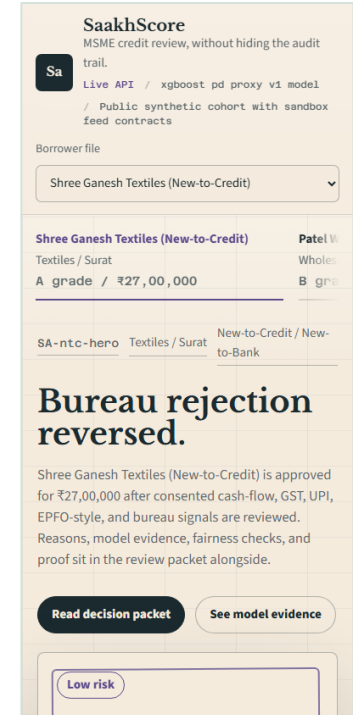
## Wireframes/Mock diagrams of the proposed solution (optional)



### Underwriter cockpit, first viewport:

- Portfolio impact strip
- Case queue
- Decision pack — Shree Ganesh Textiles

**Traditional: Rejected vs SaakhScore: Approved**



Same flow, mobile width

## Architecture diagram of the proposed solution

**Clients** — underwriter cockpit (no-build vanilla JS, EN/हिन्दी) · Swagger /docs · curl & CI probes

### Single FastAPI service (Docker, non-root, Render) — REST API + static frontend

|                                                        |                                                     |                                             |
|--------------------------------------------------------|-----------------------------------------------------|---------------------------------------------|
| scoring.py · 5-pillar policy + versioned policy object | ml.py + xgb_pd_model.py · champion + exact TreeSHAP | pd_model.py · calibrated logistic fallback  |
| feed_ingestion.py · AA/GST/UPI/EPFO + enforced consent | auth.py + rate_limit.py · bearer RBAC               | whatif.py + rails.py · what-if + OCEN offer |
| screening.py · advisory negative-registry              | audit_log.py · hash-chained tamper-evident log      | validation.py · AUC/Gini/KS/PSI/stability   |
| portfolio.py · impact + governance (PII-redacted)      | recalibration.py · sandbox readiness checks         | pilot_metrics.py · NTC/NTB lift targets     |

### Offline training — backend/model\_training/

dataset → artifact pipeline, hash-verified, never imported by the serving app;  
19-run experiment registry

### Ops — GitHub Actions CI (150 tests + container job)

render.yaml Blueprint deploy · /health/ready with commit + artifact integrity ·  
strict CSP, zero external requests

## Technologies to be used in the solution

Python 3.12

FastAPI

Pydantic

pytest + Playwright

vanilla HTML/CSS/JS

XGBoost (monotonic, offline)

exact TreeSHAP

Docker (non-root)

Render + render.yaml

GitHub Actions CI

### Stage-2 targets (already contract-ready in the repo):

- Live IDBI sandbox credentials and repayment labels → IDBI-calibrated champion
- SSO/KMS, durable audit storage, production monitoring
- Shipped now: feed contracts, dated 12-month outcome schema, chronological development/calibration/OOT readiness gates, fail-closed promotion, validation metrics, fairness support checks, pilot KPIs, optional Bedrock memo fallback



## Estimated implementation cost (optional)

### Public prototype (today)

Single container on Render free tier · synthetic demo data · no managed database needed · effectively zero infra cost

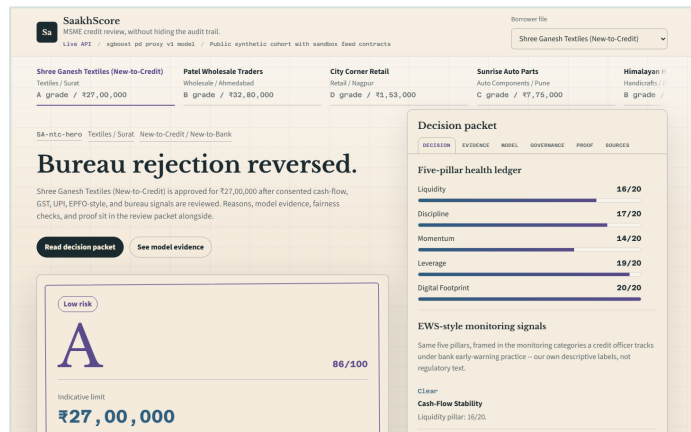
### Stage-2 pilot

Cloud container service + managed Postgres + AWS Bedrock (memo) + monitoring + authenticated sandbox API integration — modest, usage-scaled

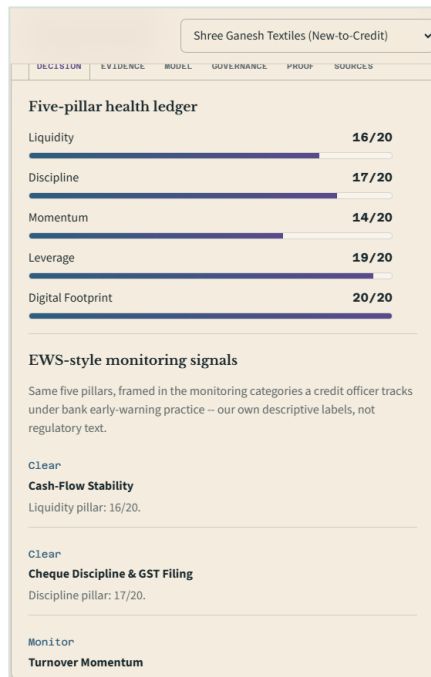
### Cost control by design

Deterministic memo fallback (no mandatory LLM spend) · dependency-light ML · single-service architecture keeps demo and pilot resilient

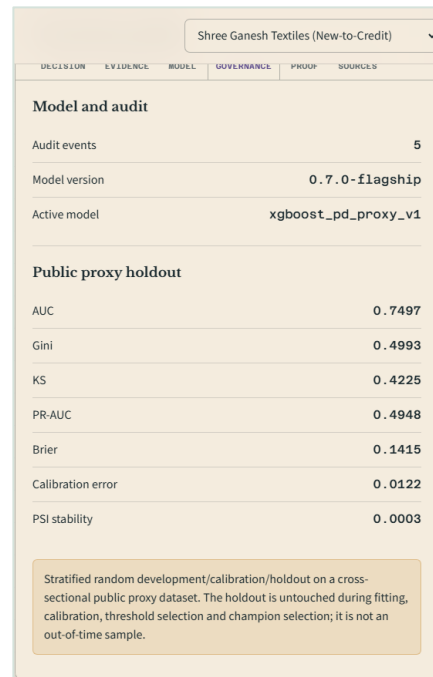
## Snapshots of the prototype



Cockpit first viewport — portfolio impact, case queue, grade A, Rs 27,00,000 limit, Traditional Rejected / SaakhScore Approved



Decision pack — pillars, reasons, Shapley, memo



Governance rail — sources, guardrails, audit, fairness

## Prototype Performance report/Benchmarking

**150 automated tests passing** (127 backend + 9 frontend unit + 14 Playwright e2e) + non-root container build/runtime/fail-closed CI job

**Public proxy holdout (real consumer default outcomes):** ROC-AUC 0.7497 (95% CI 0.7314–0.7678), Gini 0.4993, KS 0.4225, Brier 0.1415, ECE 0.0122 on 4,500 untouched rows — cross-sectional random holdout, reproducible via `train_pd_model.py`

**Real small-business benchmark (GET /model/sme-benchmark):** v2 champion selected across 418,947 resolved real SBA 7(a) loans at natural base rates (197,716-loan train split); **true out-of-time FY2017–19: ROC-AUC 0.9623, KS 0.82** on 114,770 loans, plus a 257k-loan recession stress cohort (ROC-AUC 0.9255). US proxy domain, disclosed — not an IDBI calibration

- Coverage: scoring, validation, grade boundaries, NTC reversal, hash-chain tamper detection, consent enforcement, auth/RBAC, Shapley invariants, feed mapping, portfolio, governance, API
- Browser-verified: no console errors, no horizontal overflow, desktop + mobile

Shree Ganesh Textiles (New-to-Credit)

| DECISION | EVIDENCE | MODEL | GOVERNANCE | PROOF | SOURCES |
|----------|----------|-------|------------|-------|---------|
|----------|----------|-------|------------|-------|---------|

Real-outcome benchmark · v2

SBA 7(a) FOIA loan-level records (official U.S. SBA, public domain) at natural base rates. FY2010-16 train/calibration/holdout; FY2017-19 true out-of-time; FY2005-07 recession stress

| Split                        | AUC                         | KS     | n        |
|------------------------------|-----------------------------|--------|----------|
| Holdout (FY2010-16)          | 0.9634<br>[0.9614 - 0.9662] | 0.8116 | 45,626   |
| True out-of-time (FY2017-19) | 0.9623<br>[0.9605 - 0.9638] | 0.8191 | 1,14,770 |
| Recession stress (FY2005-07) | 0.9255                      | 0.742  | 2,57,465 |

DeLong-tested baselines

| Baseline (identical protocol) | OOT AUC | DeLong vs v2 |
|-------------------------------|---------|--------------|
| Logistic Same Features        | 0.8571  | z = 59.002   |
| V1 Recipe Blanket Monotone    | 0.9097  | z = 44.339   |

E3/E3b: blanket constraints cost 5.2pp OOT AUC because real SBA term-risk is non-monotone at the short end; freeing term while keeping every defensible constraint beat the unconstrained model out-of-time (DeLong z=-8.6) and under stress.

Artifact integrity: **pass** – served model hashes match the committed evaluation.

## Additional Details/Future Development (if any)

1. Connect authenticated IDBI sandbox AA/GST/UPI/EPFO feeds to the implemented /sandbox/score contract
2. Validate dated repayment outcomes via /sandbox/pilot-readiness; retrain champion/challenger offline; promote only after /deployment/readiness passes
3. Benchmark SHAP-backed GBM output against the transparent scorecard (model-provider switch already shipped)
4. Enable AWS Bedrock memo generation in the pilot environment (deterministic fallback already present)
5. Move policy to PD-threshold-based policy-v2; move RBAC/audit storage to IDBI SSO + persistent multi-instance storage (docs/SECURITY\_COMPLIANCE.md tracks implemented vs deferred)
6. Track NTC/NTB approval lift, decision-time reduction, early-NPA guardrail and diversification with the implemented pilot metrics — real numbers, not targets

## Provide links to your:

- GitHub Public Repository → <https://github.com/bansalbhunesh/id>
- Demo Video Link (3 Minutes) → <https://youtu.be/0I5BY6DKB1E>
- Final Product Link → <https://id-ysm9.onrender.com>

### Also in the repository:

Animated walkthrough: docs/demo.gif · Model card: MODEL\_CARD.md · Benchmark analysis: docs/research/BENCHMARK\_REPORT.md · Security: SECURITY.md + docs/SECURITY\_COMPLIANCE.md · Pilot runbook: docs/PILOT\_RUNBOOK.md

Demo bearer keys for the live sandbox console are documented in the README (demo-scoped by design).





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# Thank you!